VIKHYAT CHAUHAN

+1-201-616-6373 | vikhyat.chauhan@gmail.com | LinkedIn | GitHub | Portfolio

SUMMARY

Production AI engineer who ships — TensorRT inference pipelines on FDA-regulated hardware, ROS 2 simulation stacks, and LangGraph agentic systems, all with measurable results. I close the gap between research and deployment: I'll take your hardest AI/ML bottleneck from prototype to production-grade, quantified, and compliant.

SKILLS

- **AI & LLM Engineering**: RAG (two-stage retrieval, BGE embeddings, cross-encoder reranking), LangGraph, LangChain, LlamaIndex, LLM fine-tuning (LoRA), LLM-as-judge, agentic workflows, semantic caching, OpenAI APIs, Llama 3, Qwen, FAISS
- **ML & Inference**: PyTorch, TensorFlow, TensorRT (ONNX export, FP16/INT8 precision tuning, custom CUDA/C++ plugins, layer fusion), HuggingFace Transformers, scikit-learn, MedCLIP, Conv1D Autoencoders
- **Cloud & Infrastructure**: GCP (Cloud Run, GKE, Vertex AI), AWS (EC2, Lambda, S3, EKS), Kubernetes, Docker, FastAPI, Spring Boot, CI/CD (Jenkins, SonarQube), MongoDB, RabbitMQ, Kafka, MQTT, Microservices, REST, regulated medical SDLC (IEC 62304, 21 CFR Part 11)
- **Languages, Frontend & Embedded**: Python, C++, TypeScript, SQL, CUDA; React, Flutter, Astro; embedded firmware (ESP32, Arduino), PCB design; Linux, Git

PROFESSIONAL EXPERIENCE

Virginia Polytechnic Institute and State University — **2024 - 2026**
Graduate AI/ML Researcher — *Blacksburg, VA*

- Designed and experimentally validated the Conflict Architecture (CA) framework for brain-inspired UAV navigation (NSF Award #2204780, PI: J.M. Paul), achieving a novel simultaneous 34.6% reduction in compute energy and elapsed time over baseline through parallel conflict resolution — a result with strong statistical significance ($p<0.0001$, $n=1{,}000$ runs, 3 real-world-mapped environments)
- *Independent project (personal extension of thesis infrastructure):* Architected a LangGraph + Llama 3 multi-agent pipeline for grammar-constrained SDF XML generation, enabling procedural authoring of 100+ simulation environments — applicable to simulation and game-world tooling.

GE HealthCare — **2022 - 2024**
Software Engineer II — *Bengaluru, IN*

- Led the technical architecture migration from a legacy C++ monolith to Kubernetes-based Java/Spring Boot services, reducing MRI application runtime by 80% by designing and implementing image normalization and segmentation algorithms in C++.
- Profiled and optimized a Swin Transformer segmentation model end-to-end - ONNX export, FP16 precision tuning, custom CUDA/C++ plugin authoring for unsupported ops, and layer fusion via TensorRT builder API - achieving 20% throughput uplift on FDA-regulated hardware enabling real-time intraoperative use.
- Developed a multimodal MRI report generation system (MedCLIP, PyTorch, TensorRT) fusing scan sequences with patient history, achieving quantifiable impact with a 6.5% improvement in diagnostic classification F1 over baselines.
- Introduced automated quality gates and CI/CD pipelines (Jenkins, SonarQube) for production deployment and optimization, reducing release defects by 60% while maintaining 0 patient-safety regressions across 12 quarterly releases.

The Next Move Electronics (TNM) — **2020 - 2022**
Co-Founder & Sole Technical Lead — *Delhi, IN*

- Founded and bootstrapped (~$10K self-funded) a home-automation hardware startup, owning every layer from PCB design and firmware to cloud backend and a Flutter/Alexa mobile app; shipped to 100+ homes before COVID supply-chain disruption ended the venture.
- Co-invented and patented a master-slave microcontroller communication protocol enabling single-chip control of multiple devices - cutting final-product Bill of Materials (BOM) cost 80% and directly enabling first design-win customers.
- Architected an AWS-backed microservice backend (EC2, Lambda, RabbitMQ, MongoDB, MQTT broker), scaling from 0 to 1000+ devices at 99.9% uptime, and cutting field bug reports 98% via a structured hardware-software validation workflow built as the sole engineer.

PROJECTS

Professional RAG Pipeline

- Built a two-stage retrieval pipeline (BGE-base embeddings → MS-MARCO Cross-Encoder reranker) evaluated with Hit@K and MRR metrics; LLM-as-judge scoring showed consistent faithfulness gains over single-stage retrieval across 50+ test queries - implemented from scratch without framework abstractions

- Implemented SHA-256 query hashing for semantic caching, reducing LLM API overhead by 30%; added per-query token/cost tracking for full inference observability.
- Deployed as a containerized microservice on Google Cloud Run with latency instrumentation by stage and grounded generation logic to eliminate model hallucinations.

Web Automation Agent

- Built a multi-step agentic browser automation system using LangGraph orchestration and Browser Use for DOM interaction, powered by a locally-quantized Qwen model on RTX 5060 Ti - no cloud API dependency.
- Designed novel agentic workflows with stateful task graphs, retry logic, conditional branching, and form-filling across multi-page web workflows.
- Achieved ~60s end-to-end task completion with sub-300ms first-token latency on local inference.

Physiological Stress Classifier

- Developed a Conv1D Autoencoder pipeline to compress high-dimensional physiological data (BVP, EDA, EMG, TEMP) from wearable sensors into compact unsupervised representations.
- Engineered a multimodal classification system on the WESAD dataset, achieving 83.5% accuracy in three-class emotion detection across 15 subjects.
- Implemented preprocessing and normalization for heterogeneous sensor sampling rates, transforming raw bio-signals into features for real-time stress monitoring.

CANavigator — Brain-Inspired Drone Navigation Framework

- Architected a modular ROS 2 + Gazebo Harmonic simulation stack with procedural city-grid and Perlin-noise NFZ generators, deterministic seeded RNG for bit-exact replay, and a physics-accurate DJI FlyCart 30 dynamics model (2nd-order actuator latency, aerodynamic drag, Ornstein-Uhlenbeck wind gusts).
- Implemented a dual energy model tracking CPU computational cost alongside motion energy (208.9 J/m Energy Per Meter(EPM) baseline), with per-strategy CSV telemetry across 8 metrics including compute latency, zone violations, and event deadline miss rates.
- Designed a fair head-to-head benchmarking protocol — all four planners (APE1/APE2/APE3/CA) navigate identical procedurally generated arenas with shared event sequences, with only runs where all strategies reach the target counted toward the 1,000-run corpus.

EDUCATION

Virginia Polytechnic Institute and State University — **Aug 2024 - May 2026**
M.S., Computer Engineering (Thesis Track) — *Blacksburg, VA*

- **Achievements:** Defended M.S. Thesis, Brain-Inspired Drone Navigation Using the Conflict Architecture — 4.0 GPA
- **Coursework:** CS5424: Advanced Machine Learning, CS5465: Applications of Machine Learning, ECE5504: Computer Architecture

Uttarakhand Technical University — **Jul 2016 - Dec 2020**
B.E., Electronics and Communication Engineering — *Uttarakhand, IN*

- **Achievements:** Graduated with First Division

PUBLICATIONS

- V. Chauhan, *Brain-Inspired Drone Navigation Using the Conflict Architecture*, M.S. Thesis, Virginia Tech, 2026.
- D. Dave, G. Sawhney, V. Chauhan, "Multi-Agent Stock Prediction Systems: Machine Learning Models, Simulations, and Real-Time Trading Strategies," arXiv:2502.15853, 2025.
- V. Chauhan et al., *Master-Slave Microcontroller Communication System for Home Automation*, The Next Move Electronics — Patent No. 202111060973 (Granted), 2021.

AWARDS & RECOGNITION

- **GE Impact Award:** Recognized for independently modernizing a critical legacy MR imaging application to production microservices, directly improving clinical software delivery at scale
- **NSF-Funded Graduate Research Assistant:** Supported on NSF Award #2204780 (CCF: Small: Paradox and Brain-inspired Computer Architecture, PI: JoAnn M. Paul); contributed significantly to thesis research on brain-inspired autonomous systems
- **Honor Society of Phi Kappa Phi:** Inducted, Virginia Tech Chapter (2026); top 10% of Master's graduate students across all disciplines
- **Omicron Delta Kappa National Leadership Honor Society:** Selected for membership, Alpha Omicron Circle, Virginia Tech (2026 Cohort); recognizes achievement in academics, research, and service